UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PROGRAMMING I
COURSE CODE	CSC402
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (10 MARKS)

Answer ALL questions.

1. Which statement immediately exits a loop?
   A. cin
   B. return 0
   C. continue
   D. break

2. Given the declaration, what is displayed?

```
char word[6] = {'p', 'r', 'i', 'n', 't', '\0'};
cout << word[3];
```

   A. i
   B. t
   C. p
   D. n

3. Which phase of the program development life cycle defines what the program must do?
   A. Testing
   B. Maintenance
   C. Analysis
   D. Coding

4. Which keyword prevents a variable's value from being changed after initialisation?
   A. void
   B. static
   C. break
   D. const

5. Which of the following declarations are valid?
   i. int a[5];
   ii. int b;
   iii. int c[3] = {1, 2, 3};
   iv. int d[];
   A. ii and iv
   B. i and iii
   C. i and ii
   D. iii and iv

6. Which part of a for loop is executed after each iteration?
   A. The update expression
   B. The condition only
   C. The initialisation
   D. The loop body twice

7. What is displayed by the following code?

```
for (int i = 1; i <= 4; i++)
    cout << i * 2;
```

   A. 2468
   B. 1357

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C. 246
D. 1234

8. What does a compiler do?
A. Stores only integer values
B. Translates the whole source program before execution
C. Draws flowcharts automatically
D. Executes one source statement at a time

9. Which of the following are valid identifiers?
i. total_score
ii. 2total
iii. _count
iv. int
A. i and iii
B. ii and iv
C. i and ii
D. iii and iv

10. Which operator reads data from cin?
A. =
B. %
C. >>
D. <<

PART B (60 MARKS)

QUESTION 1

This question tests the program development life cycle.

a) Define source code and object code.

(2 marks)

b) Describe THREE stages of the program development life cycle.

(3 marks)

QUESTION 2

A program reads a student's score from 0 to 100 and displays the grade shown in Table 1.

Table 1. Grade Scale

Score	Grade
80-100	A
60-79	B
50-59	C
40-49	D
0-39	F

a) Draw a flowchart for the grade classification.

(8 marks)

b) Write pseudocode for the algorithm and convert it into a complete C++ program.

(5 marks)

QUESTION 3

This question tests operator precedence and assignment.

a) Evaluate the following expressions using the declarations in the code block.

```
int a = 12, b = 5;
double c = 2.5;
```

(6 marks)

b) Write C++ statements to declare two integers a and b, read their values, and display their sum, difference, product and remainder.

(5 marks)

QUESTION 4

Study the following array program segment:

```
int a[5] = {2, 4, 6, 8, 10};
int s = 0;
for (int i = 0; i < 5; i++) {
    s += a[i];
}
cout << s;
```

a) Trace the program and state the output.

(6 marks)

b) Rewrite the segment using a do...while loop that sums only values greater than 5.

(9 marks)

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QUESTION 5

A sports club records the sport names and medal counts in Table 1.

a) Declare two parallel arrays for the five sports in Table 1 and initialise them with those values.

Table 2. Sport and Medal Records

Sport	Medals
Swimming	8
Badminton	12
Athletics	7
Cycling	5
Hockey	9

(6 marks)

b) Write a program segment to display each sport with its medals, then calculate the total and average medals.

(10 marks)

PART C (30 MARKS)

QUESTION 1

A lecturer records the marks of ten students in an array.

Write a complete C++ program that reads ten marks, displays the average, the highest mark and its index, and counts how many marks are above the average.

(15 marks)

QUESTION 2

A bookstore records up to five books. Table 1 shows an example of the fields.

Table 3. Book Record Fields

Title	Price (RM)	Quantity
Book A	45.00	3
Book B	30.00	5
Book C	55.00	2

Define a struct Book. Write a complete C++ program that reads the records, calculates the total inventory value, and displays the title of the most expensive book.

(15 marks)

END OF QUESTION PAPER

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